

Chapter 01: Introduction to Differential Equations & Initial-Value Problems

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering

Course: Applied Ordinary Differential Equations (ENGR 213)

Textbook:

Advanced Engineering Mathematics

(7th Edition) by Dennis G. Zill — Chapter 1 (§1.1, §1.2, §1.3)

1. Executive Overview & First-Principles Philosophy

In engineering and applied physics, fundamental laws rarely state the value of a physical state directly. Instead, nature governs the universe through **rates of change**:

- Newton's Second Law of Motion relates acceleration (the second derivative of position) to net force: $\sum F = m \frac{d^2x}{dt^2}$.
- Fourier's Law of Heat Conduction states that heat flux is proportional to the spatial temperature gradient: $q = -k \frac{\partial T}{\partial x}$.
- Faraday's and Kirchhoff's Laws describe electrical circuits in terms of the rate of change of charge and current: $L \frac{di}{dt} + Ri = E(t)$.

A **Differential Equation (DE)** is any mathematical equation containing the derivatives of one or more unknown functions (dependent variables) with respect to one or more independent variables.

Solving a differential equation represents the inverse process of differentiation: starting from localized rate relationships and geometric constraints, determine the global function $y(x)$ that describes the state of the system across its domain.

2. Mathematical Framework & Analytical Mechanics

2.1 Classification of Differential Equations

Every differential equation encountered in engineering mathematics is categorized across three foundational characteristics: **Type**, **Order**, and **Linearity**.

Differential Equations Classification

- └ 1. By Type
 - └ Ordinary Differential Equation (ODE): Single independent variable
 - └ Partial Differential Equation (PDE): Two or more independent variables
- └ 2. By Order
 - └ Highest derivative present in the equation: $d^n y/dx^n$
- └ 3. By Linearity
 - └ Linear: First degree in $y, y', y'', \dots$, coefficients depend only on x
 - └ Nonlinear: Powers of y or derivatives, cross-products, transcendental functions

A. Classification by Type

1. **Ordinary Differential Equation (ODE):** An equation containing only ordinary derivatives of one or more dependent variables with respect to a **single** independent variable:

$$\frac{dy}{dx} + 5y = e^x, \quad \frac{d^2x}{dt^2} + \frac{dx}{dt} + 6x = 0, \quad \frac{dx}{dt} + \frac{dy}{dt} = 2x + y$$

2. **Partial Differential Equation (PDE)**: An equation involving partial derivatives of one or more dependent variables with respect to **two or more** independent variables:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad (\text{Laplace's Eq.}), \quad k \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t} \quad (\text{Heat Eq.})$$

B. Classification by Order

- The **order** of an ODE is the order of the **highest derivative** present in the equation.

Example 1

: $\frac{d^2 y}{dx^2} + 5 \left(\frac{dy}{dx} \right)^3 - 4y = e^x$ is **Second-Order** (highest derivative is $\frac{d^2 y}{dx^2}$, despite the first derivative being cubed).

Example 2

: $4xy''' + 2y'' - (\sin x)y' + y = 0$ is **Third-Order**.

- Normal Form**: An n -th order ODE can be expressed in general implicit form:

$$F(x, y, y', y'', \dots, y^{(n)}) = 0$$

Whenever algebraically possible, we isolate the highest derivative to express the ODE in **normal form**:

$$\frac{d^n y}{dx^n} = f(x, y, y', \dots, y^{(n-1)})$$

For first-order ODEs, the normal form is simply $\frac{dy}{dx} = f(x, y)$. For second-order ODEs, it is $\frac{d^2 y}{dx^2} = f(x, y, y')$.

C. Classification by Linearity

An n -th order ordinary differential equation is defined as **linear** if it can be written strictly in the form:

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

The Two Non-Negotiable Tests for Linearity:

- First-Degree Rule**: The dependent variable y and all its derivatives $y', y'', \dots, y^{(n)}$ must appear strictly to the first power (exponent 1). There can be no terms such as y^2 , $\sqrt{y}$, or $(y')^3$.
- Coefficient Rule**: The coefficients $a_0(x), a_1(x), \dots, a_n(x)$ and the non-homogeneous driving term $g(x)$ must depend **solely on the independent variable x** (or be constant). They cannot depend on y or any derivatives of y .
- No Nonlinear Compositions**: No transcendental functions of the dependent variable may appear (e.g., $\sin(y)$, $\cos(y)$, e^y , $\ln(y)$).
- No Cross-Products**: No products of the dependent variable with its derivatives may appear (e.g., $y \cdot y'$, $y' \cdot y''$).

Differential Equation	Order	Linear?	Rationale / Non-Linear Term
$(y - x)dx + 4xdy = 0$	1	Linear in y	Rewrites as $4x \frac{dy}{dx} + y = x$. Linear coefficients in x .
$y'' - 2y' + y = \sin x$	2	Linear	Constant coefficients; y and derivatives to power 1.
$x^3y''' + xy' - 5y = e^x$	3	Linear	Coefficients depend only on x ; linear in y .
$(1 - y)y' + 2y = e^x$	1	Nonlinear	Term $-yy'$ has dependent variable multiplying its derivative.
$\frac{d^2y}{dx^2} + \sin y = 0$	2	Nonlinear	Transcendental function $\sin y$ depends on y .
$\frac{d^4y}{dx^4} + y^2 = 0$	4	Nonlinear	Term y^2 is second-degree in y .
$\left(\frac{dy}{dx}\right)^2 + y = x$	1	Nonlinear	Derivative is squared $(y')^2$.

2.2 Solutions of Differential Equations

A. Definition of a Solution

A function ϕ defined on an interval I and possessing at least n derivatives that are continuous on I is called a **solution** of an n -th order ODE on I if, when $y = \phi(x)$ and its derivatives are substituted into the equation, it reduces the equation to an identity for all $x \in I$:

$$F\left(x, \phi(x), \phi'(x), \dots, \phi^{(n)}(x)\right) \equiv 0 \quad \forall x \in I$$

B. Interval of Definition (Domain of Solution)

The interval I is variously called the **interval of definition**, the **interval of existence**, the **domain of the solution**, or the **interval of validity**. The interval I must be continuous: it can be an open interval (a, b) , a closed interval $[a, b]$, or an infinite interval $(-\infty, \infty)$. A solution cannot bridge across a point of discontinuity or singularity.

C. Explicit vs. Implicit Solutions

1. **Explicit Solution:** A solution in which the dependent variable is expressed entirely in terms of the independent variable and constants:

$$y = \phi(x)$$

Example

: $y = ce^{2x} + 4$ is an explicit solution of $y' - 2y = -8$.

2. **Implicit Solution:** A relation $G(x, y) = 0$ is said to be an implicit solution of an ODE on interval I , provided that there exists at least one function $y = \phi(x)$ that satisfies the relation as well as the differential equation on I .

Example

: $x^2 + y^2 - 25 = 0$ is an implicit solution of $\frac{dy}{dx} = -\frac{x}{y}$ on the open interval $(-5, 5)$. Implicit differentiation yields:

$$\frac{d}{dx}[x^2 + y^2 - 25] = 2x + 2y \frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{x}{y}$$

This implicit relation defines two distinct explicit solutions:

$$\phi_1(x) = +\sqrt{25 - x^2} \quad \text{and} \quad \phi_2(x) = -\sqrt{25 - x^2}, \quad x \in (-5, 5)$$

D. Families of Solutions & Singular Solutions

- **n -Parameter Family of Solutions:** When solving an n -th order ODE $F(x, y, y', \dots, y^{(n)}) = 0$, we integrate n times, yielding a solution containing n arbitrary constants:

$$G(x, y, c_1, c_2, \dots, c_n) = 0$$

This is called a **general solution** or **n -parameter family**.

- **Particular Solution:** A solution free of arbitrary parameters, obtained by choosing specific numerical values for $c_1, \dots, c_n$ (usually enforced by initial conditions).
- **Singular Solution:** A solution that cannot be obtained by any choice of the parameters $c_1, \dots, c_n$ in the family.

Textbook Classic

: For the first-order nonlinear ODE $(y')^2 - xy' + y = 0$, the one-parameter family is the family of straight lines:

$$y = cx - c^2$$

However, the parabola $y = \frac{1}{4}x^2$ is also a solution:

$$y' = \frac{1}{2}x \implies \left(\frac{1}{2}x\right)^2 - x\left(\frac{1}{2}x\right) + \frac{1}{4}x^2 = \frac{1}{4}x^2 - \frac{1}{2}x^2 + \frac{1}{4}x^2 = 0 \equiv 0$$

Because no choice of constant c in $y = cx - c^2$ can produce the quadratic curve $y = \frac{1}{4}x^2$, this parabola is a **singular solution**. Geometrically, it is the envelope of the family of tangent lines.

- **Trivial Solution:** If $y \equiv 0$ identically satisfies an ODE (common in homogeneous linear equations), it is called the **trivial solution**.

2.3 Initial-Value Problems (IVPs) & The Picard-Lindelöf Theorem

A. Mathematical Definition of an IVP

An **Initial-Value Problem** seeks a solution $y(x)$ to an ODE on an interval I containing x_0 , subject to side conditions prescribed at a **single point** x_0 :

- **First-Order IVP:**

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0$$

- **Second-Order IVP:**

$$\frac{d^2y}{dx^2} = f(x, y, y'), \quad y(x_0) = y_0, \quad y'(x_0) = y_1$$

- **n -th Order IVP:**

$$\frac{d^ny}{dx^n} = f(x, y, y', \dots, y^{(n-1)}), \quad y(x_0) = y_0; y'(x_0) = y_1; \dots; y^{(n-1)}(x_0) = y_{n-1}$$

Geometrically, for a first-order ODE, the initial condition $y(x_0) = y_0$ picks out a single integral curve passing through the specific Cartesian coordinate point (x_0, y_0) . For a second-order ODE, $y(x_0) = y_0$ specifies the starting position, and $y'(x_0) = y_1$ specifies the initial tangent slope.

B. The Existence and Uniqueness Theorem (Theorem 1.2.1)

In engineering, we must know:

1. **Existence:** Does a mathematical model have at least one solution?
2. **Uniqueness:** Is the solution unique, or does the system permit unpredictable branching?

Theorem 1.2.1 (Existence of a Unique Solution — Picard-Lindelöf): Let R be a rectangular region in the xy -plane defined by:

$$R = (x, y) \in \mathbb{R}^2 \mid a \leq x \leq b, c \leq y \leq d$$

containing the interior point (x_0, y_0) . If both:

1. $f(x, y)$ is **continuous** on R , and
2. $\frac{\partial f}{\partial y}$ is **continuous** on R ,

then there exists some open subinterval $I_0 = (x_0 - h, x_0 + h)$ contained in $[a, b]$, and a **unique function** $y(x)$ defined on I_0 that satisfies the IVP:

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0$$

Condition 1: $f(x, y)$ is continuous on R

--> Existence of a solution is **GUARANTEED**

Condition 2: $\partial f / \partial y$ is continuous on R

--> **UNIQUENESS** of the solution is **GUARANTEED**

If Condition 2 fails at (x_0, y_0)

--> Solutions may exist, but **UNIQUENESS FAILS!**

C. Exhaustive Analysis of Theorem 1.2.1 Failure Modes

Case Study 1: Complete Failure of Uniqueness ($y' = xy^{1/2}; y(0) = 0$)

Consider the first-order IVP:

$$\frac{dy}{dx} = xy^{1/2}, \quad y(0) = 0$$

- Step 1 ($f(x, y)$): $f(x, y) = x\sqrt{y}$. This function is continuous everywhere on the half-plane $y \geq 0$. Since $(0, 0)$ is on the boundary, existence is guaranteed.
- Step 2 ($\partial f / \partial y$):

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} [xy^{1/2}] = \frac{x}{2\sqrt{y}}$$

Evaluating at the initial condition point $(x_0, y_0) = (0, 0)$:

$$\left. \frac{\partial f}{\partial y} \right|_{(0,0)} = \frac{0}{2\sqrt{0}} \implies \text{UNDEFINED (division by zero!)}$$

Because $\frac{\partial f}{\partial y}$ is discontinuous across the line $y = 0$, **uniqueness is NOT guaranteed**.

- Analytical Consequence: Separating variables:

$$\frac{dy}{y^{1/2}} = x dx \implies 2y^{1/2} = \frac{x^2}{2} + C \implies y(x) = \left(\frac{x^2}{4} + \frac{C}{2} \right)^2$$

Applying $y(0) = 0 \implies C = 0 \implies y_1(x) = \frac{x^4}{16}$. However, notice that $y_2(x) \equiv 0$ is also a completely valid solution satisfying $y(0) = 0$. Furthermore, we can construct infinitely many piecewise solutions:

$$y(x) = \begin{cases} 0, & x < c \\ \frac{1}{16}(x^2 - c^2)^2, & x \geq c \end{cases}$$

Through the single point $(0, 0)$, infinitely many distinct integral curves emerge!

Case Study 2: Existence and Uniqueness Guaranteed Everywhere Except Vertical Lines ($y' = \frac{y}{x}$)

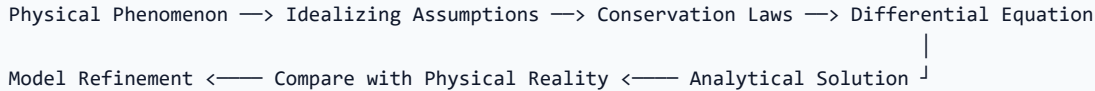
For the ODE $\frac{dy}{dx} = \frac{y}{x}$:

- $f(x, y) = \frac{y}{x}$ is continuous for all $x \neq 0$.
- $\frac{\partial f}{\partial y} = \frac{1}{x}$ is continuous for all $x \neq 0$.
- Conclusion: For any initial condition (x_0, y_0) with $x_0 \neq 0$, Theorem 1.2.1 guarantees a unique solution. Across the y -axis ($x = 0$), existence and uniqueness guarantees fail. Indeed, general solution is $y = cx$. At $(0, 0)$, all lines $y = cx$ pass through (infinitely many solutions). At $(0, 1)$, no solution passes through (zero solutions).

3. Mathematical Modeling & First-Principles Derivations

In §1.3 of the textbook, Dennis Zill establishes that formulating a mathematical model follows a four-step engineering cycle:

1. **Identify Variables:** Formulate dependent variables representing state quantities (mass, temperature, voltage) and independent variables (time, distance).
2. **Apply Physical Laws:** Enforce conservation principles (conservation of mass, energy, momentum, charge).
3. **Formulate the DE:** Express empirical relations (rates of change proportional to driving forces).
4. **Solve, Validate & Interpret:** Obtain analytical solutions, compare against experimental data, and verify physical limits ($t \rightarrow \infty$).



Model 3.1: Population Dynamics & The Malthusian Law

- **Physical Axiom:** In an unrestricted environment with abundant nutrients and space, the rate of population growth is directly proportional to the total population present at that instant.
- **Mathematical Derivation:** Let $P(t)$ be the population at time t :

$$\frac{dP}{dt} \propto P(t) \implies \frac{dP}{dt} = kP$$

where k is the constant of proportionality (birth rate minus death rate).

- **Step-by-Step Analytical Solution:**

$$\frac{dP}{P} = k, dt \implies \ln |P| = kt + C_1 \implies P(t) = Ce^{kt}$$

Applying initial condition $P(0) = P_0$:

$$P_0 = Ce^0 \implies C = P_0 \implies P(t) = P_0 e^{kt}$$

- **Limitation:** As $t \rightarrow \infty$, $P(t) \rightarrow \infty$. Real populations experience space/food limitations, leading directly to the nonlinear logistic equation (Chapter 2, §2.8).

Model 3.2: Radioactive Decay & Half-Life Radiocarbon Dating

- **Physical Axiom:** The rate at which the nuclei of a radioactive substance decay is directly proportional to the remaining number of nuclei (or mass $A(t)$) at that time.
- **Mathematical Formulation:**

$$\frac{dA}{dt} = -kA, \quad k > 0$$

The negative sign explicitly represents loss of mass over time.

- **Derivation of Half-Life ($t_{1/2}$):** The solution is $A(t) = A_0 e^{-kt}$. The half-life is the time required for half the original atoms to disintegrate:

$$A(t_{1/2}) = \frac{1}{2}A_0 \implies A_0 e^{-kt_{1/2}} = \frac{1}{2}A_0 \implies e^{-kt_{1/2}} = \frac{1}{2}$$

Taking the natural logarithm:

$$-kt_{1/2} = \ln\left(\frac{1}{2}\right) = -\ln 2 \implies t_{1/2} = \frac{\ln 2}{k} \approx \frac{0.693147}{k}$$

- **Carbon-14 Dating:** Carbon-14 has a half-life of $t_{1/2} \approx 5730$ years.

$$k = \frac{\ln 2}{5730} \approx 0.00012097 \text{ year}^{-1}$$

By measuring the residual ^{14}C activity ratio $A(t)/A_0$ in an organic artifact, the elapsed time since death is determined explicitly by:

$$t = -\frac{1}{k} \ln\left(\frac{A(t)}{A_0}\right)$$

Model 3.3: Newton's Law of Cooling and Warming

- **Physical Axiom:** The rate of change of the temperature $T(t)$ of a body is proportional to the difference between its temperature and the ambient temperature T_m of the surrounding medium.
- **Mathematical Formulation:**

$$\frac{dT}{dt} = -k(T - T_m), \quad k > 0$$

- Physical check: If $T > T_m$ (body is hotter than room), $(T - T_m) > 0$, so $\frac{dT}{dt} < 0$ (the body cools). If $T < T_m$ (body is colder than room), $(T - T_m) < 0$, so $\frac{dT}{dt} > 0$ (the body warms).

- **Step-by-Step Analytical Solution:**

$$\frac{dT}{T - T_m} = -k, dt \implies \ln|T - T_m| = -kt + C_1 \implies T(t) - T_m = Ce^{-kt}$$

Applying initial temperature $T(0) = T_0$:

$$T_0 - T_m = C \implies T(t) = T_m + (T_0 - T_m)e^{-kt}$$

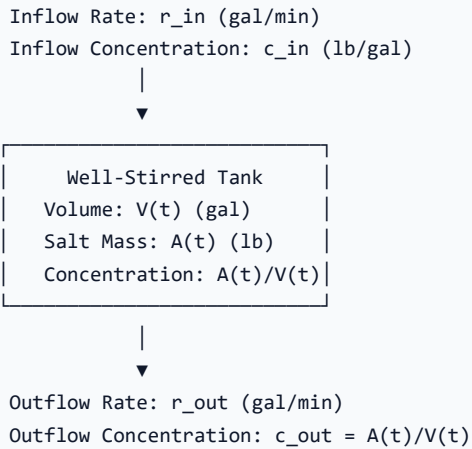
- **Asymptotic Behavior:**

$$\lim_{t \rightarrow \infty} T(t) = T_m + (T_0 - T_m)(0) = T_m$$

The body asymptotically approaches thermal equilibrium with its environment.

Model 3.4: Mixtures & Dynamic Tank Concentration Mechanics

Mixing tanks are quintessential industrial and chemical engineering systems:



- **Conservation Law:**

$$\frac{dA}{dt} = \text{Rate of chemical entering} - \text{Rate of chemical leaving} = R_{\text{in}} - R_{\text{out}}$$

- **Detailed Derivation:**

1. **Input Rate (R_{in}):**

$$R_{\text{in}} = (\text{volumetric inflow rate } r_{\text{in}}) \times (\text{solute inflow concentration } c_{\text{in}}) = r_{\text{in}} \cdot c_{\text{in}} \quad \left[\frac{\text{gal}}{\text{min}} \cdot \frac{\text{lb}}{\text{gal}} = \frac{\text{lb}}{\text{min}} \right]$$

2. **Fluid Volume Dynamics ($V(t)$):**

$$\frac{dV}{dt} = r_{\text{in}} - r_{\text{out}} \implies V(t) = V_0 + (r_{\text{in}} - r_{\text{out}})t$$

3. **Output Rate (R_{out}):** Assuming instantaneous, complete uniform mixing via an impeller, the concentration of solute leaving the tank equals the internal concentration:

$$c_{\text{out}}(t) = \frac{A(t)}{V(t)} = \frac{A(t)}{V_0 + (r_{\text{in}} - r_{\text{out}})t}$$
$$R_{\text{out}} = r_{\text{out}} \cdot c_{\text{out}}(t) = r_{\text{out}} \frac{A(t)}{V_0 + (r_{\text{in}} - r_{\text{out}})t}$$

4. **The Governing Linear ODE:**

$$\frac{dA}{dt} = r_{\text{in}}c_{\text{in}} - r_{\text{out}} \frac{A(t)}{V_0 + (r_{\text{in}} - r_{\text{out}})t}$$

In standard linear form:

$$\frac{dA}{dt} + \left(\frac{r_{\text{out}}}{V_0 + (r_{\text{in}} - r_{\text{out}})t} \right) A = r_{\text{in}}c_{\text{in}}$$

- **Two Operating Regimes:**

- **Equal Flow Rates** ($r_{\text{in}} = r_{\text{out}} = r$): Constant volume $V(t) = V_0$. The ODE has constant coefficients:

$$\frac{dA}{dt} + \frac{r}{V_0} A = r c_{\text{in}}$$

- **Unequal Flow Rates** ($r_{\text{in}} \neq r_{\text{out}}$): The volume changes linearly with time. If $r_{\text{in}} > r_{\text{out}}$, the tank fills and will overflow at $t_{\text{overflow}} = \frac{V_{\text{max}} - V_0}{r_{\text{in}} - r_{\text{out}}}$. If $r_{\text{in}} < r_{\text{out}}$, the tank empties completely at $t_{\text{empty}} = \frac{V_0}{r_{\text{out}} - r_{\text{in}}}$.

Model 3.5: Draining Tanks & Torricelli's Law

- **Physical Axiom:** Water flows through a hole of area a at the bottom of an open tank with velocity $v = \sqrt{2gh}$, where h is the water depth (Torricelli's Law of Torricellian Efflux, derived from Bernoulli's principle).
- **Volumetric Balance:** The volume of water exiting the tank in time dt is:

$$dV = -avdt = -ac\sqrt{2gh}, dt$$

where c is an empirical discharge contraction coefficient ($0 < c \leq 1$). If the tank has a cross-sectional area $A_h(h)$ at height h :

$$dV = A_h(h)dh$$

- **The Governing Nonlinear ODE:**

$$A_h(h) \frac{dh}{dt} = -ac\sqrt{2gh}$$

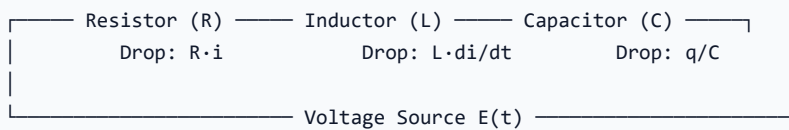
- For a cylindrical tank with constant cross-section A_w :

$$\frac{dh}{dt} = -\frac{ac\sqrt{2g}}{A_w} \sqrt{h}$$

This is a separable nonlinear ODE whose solution shows that $h(t)$ decreases quadratically, reaching zero at finite time t_{empty} .

Model 3.6: Series Electrical Circuits (LR, RC, LRC)

By Kirchhoff's Second Law (Loop Rule), the algebraic sum of all voltage drops across a closed loop equals the applied electromotive force $E(t)$:



- **Component Voltage Drop Formulas:**

- Resistor: $E_R = R \cdot i = R \frac{dq}{dt}$
- Inductor: $E_L = L \cdot \frac{di}{dt} = L \frac{d^2q}{dt^2}$
- Capacitor: $E_C = \frac{1}{C} q$

- **Governing Circuit Equations:**

1. **Series LR Circuit:**

$$L \frac{di}{dt} + Ri = E(t)$$

2. Series RC Circuit:

$$R \frac{dq}{dt} + \frac{1}{C} q = E(t)$$

Differentiating with respect to t gives the current formulation:

$$R \frac{di}{dt} + \frac{1}{C} i = \frac{dE}{dt}$$

3. Series LRC Circuit:

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = E(t)$$

Model 3.7: Falling Bodies, Gravitation & Drag Resistance

By Newton's Second Law: $m \frac{dv}{dt} = \sum F$. Taking the downward direction as positive (x downward, $v = \frac{dx}{dt} > 0$):

1. No Air Resistance:

$$m \frac{dv}{dt} = mg \implies \frac{dv}{dt} = g \implies v(t) = gt + v_0$$

2. Linear Drag (Low-Velocity / Laminar Flow, Stokes' Law): Retarding force $F_{\text{drag}} = -kv$ ($k > 0$):

$$m \frac{dv}{dt} = mg - kv \implies \frac{dv}{dt} + \frac{k}{m} v = g$$

Solving via integrating factor $\mu(t) = e^{kt/m}$:

$$v(t) = \frac{mg}{k} + \left(v_0 - \frac{mg}{k} \right) e^{-kt/m}$$

As $t \rightarrow \infty$, the exponential vanishes, yielding the **terminal velocity**:

$$v_{\text{term}} = \lim_{t \rightarrow \infty} v(t) = \frac{mg}{k}$$

3. Quadratic Drag (High-Velocity / Turbulent Flow): Retarding force $F_{\text{drag}} = -kv^2$:

$$m \frac{dv}{dt} = mg - kv^2 \implies \frac{dv}{dt} = g \left(1 - \frac{k}{mg} v^2 \right)$$

Terminal velocity occurs when acceleration $\frac{dv}{dt} = 0$:

$$v_{\text{term}} = \sqrt{\frac{mg}{k}}$$

Separating variables:

$$\int \frac{dv}{v_{\text{term}}^2 - v^2} = \frac{k}{m} \int dt \implies v(t) = v_{\text{term}} \tanh \left(\frac{gt}{v_{\text{term}}} \right)$$

4. Visual Architecture & Textbook Figure Deep Dives

Visual 4.1: Direction Fields & Lineal Elements (Textbook Fig. 2.1.1)

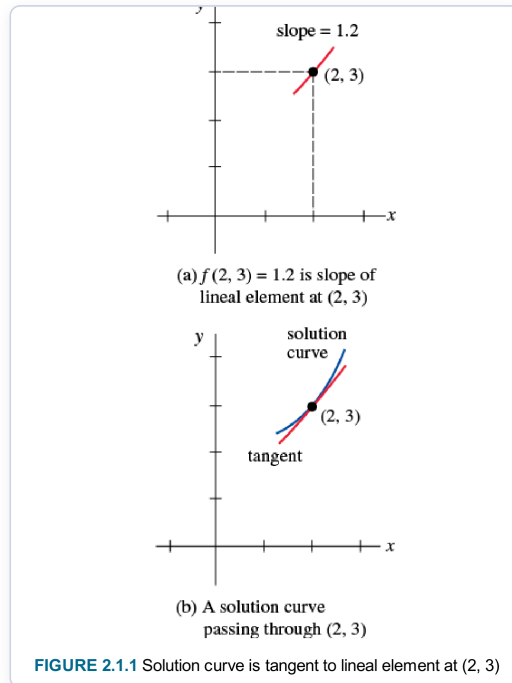


Figure 1.1: Lineal element indicating slope at point (x, y) — from Textbook 7th Ed. Chapter 2 (Fig. 2.1.1).

Pedagogical Breakdown:

1. Mathematical Representation of Slope:

- The first-order differential equation in normal form $\frac{dy}{dx} = f(x, y)$ is a direct statement that the slope of the tangent line to the solution curve at (x, y) is given by the function value $f(x, y)$.
- At any coordinate point (x_0, y_0) , we draw a short line segment (lineal element) centered at (x_0, y_0) with slope $m = f(x_0, y_0)$.

2. Geometric Interpretation of Solutions:

- A solution curve $y = \phi(x)$ is a smooth curve that passes through the direction field in such a manner that at every point $(x, \phi(x))$, the curve is strictly tangent to the lineal element.
- If a grid of lineal elements is drawn across a rectangular region, the overall pattern displays the **flow lines** or stream trajectories of all possible solutions without calculating a single integral!

5. Fully Worked Textbook & Exam Archetypes

Archetype 5.1: Rigorous Verification of an Implicit Solution

Problem: Verify that $x^2 + y^2 = 25$ is an implicit solution to the ODE $\frac{dy}{dx} = -\frac{x}{y}$ on the open interval $-5 < x < 5$, and clearly identify all explicit solutions.

Step-by-Step Baby-Step Solution:

- **Step 1: Differentiate the implicit relation with respect to x :** Applying the chain rule to the term y^2 :

$$\frac{d}{dx} [x^2 + y^2] = \frac{d}{dx} [25]$$

$$2x + 2y \frac{dy}{dx} = 0$$

- **Step 2: Solve algebraically for the derivative $\frac{dy}{dx}$.**

$$2y \frac{dy}{dx} = -2x \implies \frac{dy}{dx} = -\frac{2x}{2y} = -\frac{x}{y}$$

This matches the given differential equation identically!

- **Step 3: Determine the explicit solutions and their domain:** Solving $x^2 + y^2 = 25$ explicitly for y :

$$y^2 = 25 - x^2 \implies y = \pm \sqrt{25 - x^2}$$

This yields two distinct continuous explicit solutions:

$$\phi_1(x) = +\sqrt{25 - x^2}, \quad -5 < x < 5$$

$$\phi_2(x) = -\sqrt{25 - x^2}, \quad -5 < x < 5$$

- **Step 4: Check interval boundaries ($x = \pm 5$):** At $x = 5$ and $x = -5$, $y = 0$. The derivative $\frac{dy}{dx} = -\frac{x}{y} = -\frac{\pm 5}{0}$ becomes infinite (vertical tangent). Therefore, the endpoints $x = \pm 5$ cannot be included in the interval of definition. The domain must be the open interval $I = (-5, 5)$.

Archetype 5.2: Investigating Existence & Uniqueness via Theorem 1.2.1

Problem: For the differential equation $\frac{dy}{dx} = \frac{y^2 - 1}{x - 2}$, determine the regions in the xy -plane where:

1. At least one solution is guaranteed to exist.
2. A unique solution is guaranteed to exist.
3. Does Theorem 1.2.1 guarantee a unique solution through $(2, 3)$? Through $(0, 1)$?

Step-by-Step Baby-Step Solution:

- **Step 1: Identify $f(x, y)$ and check continuity:**

$$f(x, y) = \frac{y^2 - 1}{x - 2}$$

$f(x, y)$ is a rational function. Rational functions are continuous everywhere their denominator is non-zero. Therefore, $f(x, y)$ is continuous on the entire xy -plane **except along the vertical line $x = 2$** .

Conclusion 1

: Existence of solutions is guaranteed in any region that does not intersect the line $x = 2$.

- **Step 2: Compute $\frac{\partial f}{\partial y}$ and check continuity:** Treat x as a constant and differentiate $f(x, y)$ with respect to y :

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} \left[\frac{y^2 - 1}{x - 2} \right] = \frac{1}{x - 2} \frac{d}{dy} (y^2 - 1) = \frac{2y}{x - 2}$$

$\frac{\partial f}{\partial y}$ is also continuous everywhere except along the vertical line $x = 2$.

Conclusion 2

: Uniqueness of solutions is guaranteed in any region that does not intersect the line $x = 2$.

- **Step 3: Evaluate specific points:**

- **Point (2, 3):** Here $x = 2$. The denominator of both f and $\frac{\partial f}{\partial y}$ evaluates to zero:

$$f(2, 3) = \frac{3^2 - 1}{2 - 2} = \frac{8}{0} \quad (\text{UNDEFINED})$$

Since neither f nor $\frac{\partial f}{\partial y}$ is continuous at $(2, 3)$, Theorem 1.2.1 **guarantees NOTHING** (neither existence nor uniqueness).

- **Point (0, 1):** Here $x = 0 \neq 2$.

$$f(0, 1) = \frac{1^2 - 1}{0 - 2} = \frac{0}{-2} = 0$$

$$\left. \frac{\partial f}{\partial y} \right|_{(0,1)} = \frac{2(1)}{0 - 2} = -1$$

We can construct a rectangle R centered at $(0, 1)$ (for example, $-1 \leq x \leq 1$; $0 \leq y \leq 2$) that does not contain the line $x = 2$. Both f and $\frac{\partial f}{\partial y}$ are continuous on R .

Conclusion

: Theorem 1.2.1 **GUARANTEES** that there exists an open interval around $x = 0$ on which there is a **unique** solution passing through $(0, 1)$. (Notice that $y(x) \equiv 1$ is that unique constant solution!).

Archetype 5.3: Formulating an Industrial Two-Fluid Mixture Model

Problem: A tank initially contains 200 liters of pure water. A brine solution containing 0.5 kg of salt per liter is pumped into the tank at a rate of 4 L/min. The well-stirred mixture is pumped out at a faster rate of 6 L/min.

1. Formulate the initial-value problem for the mass of salt $A(t)$ in the tank.
2. Determine the time at which the tank is completely empty, and the valid time domain of the differential equation.

Step-by-Step Baby-Step Solution:

- **Step 1: Determine the inflow rate of salt (R_{in}):**

$$r_{\text{in}} = 4 \text{ L/min}, \quad c_{\text{in}} = 0.5 \text{ kg/L}$$

$$R_{\text{in}} = r_{\text{in}} \cdot c_{\text{in}} = (4 \text{ L/min}) \times (0.5 \text{ kg/L}) = 2.0 \text{ kg/min}$$

- **Step 2: Determine the liquid volume $V(t)$ in the tank:** The net change in volume per minute is:

$$\frac{dV}{dt} = r_{\text{in}} - r_{\text{out}} = 4 - 6 = -2 \text{ L/min}$$

Integrating with initial condition $V(0) = 200 \text{ L}$:

$$V(t) = V_0 + (r_{\text{in}} - r_{\text{out}})t = 200 - 2t \text{ liters}$$

- **Step 3: Determine the concentration and outflow rate of salt (R_{out}):**

$$\text{Salt concentration in tank } c(t) = \frac{A(t)}{V(t)} = \frac{A(t)}{200 - 2t} \text{ kg/L}$$

$$R_{\text{out}} = r_{\text{out}} \cdot c(t) = 6 \cdot \frac{A(t)}{200 - 2t} = \frac{6A(t)}{2(100 - t)} = \frac{3A(t)}{100 - t} \text{ kg/min}$$

- **Step 4: Assemble the differential equation and initial condition:**

$$\frac{dA}{dt} = R_{\text{in}} - R_{\text{out}} \implies \frac{dA}{dt} = 2 - \frac{3A}{100 - t}$$

In standard linear form:

$$\frac{dA}{dt} + \frac{3}{100 - t}A = 2, \quad A(0) = 0$$

- **Step 5: Determine the empty time and domain:** The tank empties when $V(t) = 0$:

$$200 - 2t = 0 \implies 2t = 200 \implies t_{\text{empty}} = 100 \text{ minutes}$$

The differential equation is valid on the physical time interval:

$$0 \leq t < 100 \text{ minutes}$$

At $t = 100$, the denominator $100 - t = 0$, creating a physical and mathematical singularity.

6. Comprehensive Diagnostic Traps & Exam Pitfalls

-  **Trap 1: Derivative Order vs. Polynomial Power:**

$$\left(\frac{dy}{dx}\right)^5 + 3\frac{d^2y}{dx^2} = 0$$

Mistake

: Calling this fifth-order because of the exponent 5.

Correction

: This equation is **Second-Order**, because the highest derivative present is $\frac{d^2y}{dx^2}$. Its degree in the first derivative is 5 (which makes it nonlinear).

-  **Trap 2: Variable Coefficients vs. Nonlinearity:**

$$x^4 y'' + (\cos x) y' + e^{x^2} y = \ln x$$

Mistake

: Declaring the equation nonlinear because of $\cos x$, e^{x^2} , and $\ln x$.

Correction

: This ODE is **strictly Linear**! Linearity restricts only the dependent variable y and its derivatives. The independent variable x is completely unrestricted and may appear inside any nonlinear transcendental function.

-  **Trap 3: Theorem 1.2.1 is SUFFICIENT, Not Necessary:**

- If $f(x, y)$ and $\frac{\partial f}{\partial y}$ are continuous, a unique solution is

guaranteed

.

- If either condition fails, the theorem makes **NO statement**. A unique solution

might

still exist, multiple solutions might exist, or no solution might exist! Never state that failure of Theorem 1.2.1 proves that no solution exists.

-  **Trap 4: Forgetting the Singular Solution During Separation:** When solving $\frac{dy}{dx} = y(1 - y)$, dividing by $y(1 - y)$ assumes $y \neq 0$ and $y \neq 1$. Both $y(x) \equiv 0$ and $y(x) \equiv 1$ are constant solutions. If an initial condition is $y(0) = 1$, the unique solution is simply the constant equilibrium line $y(x) = 1$, which is lost if division is performed carelessly!